

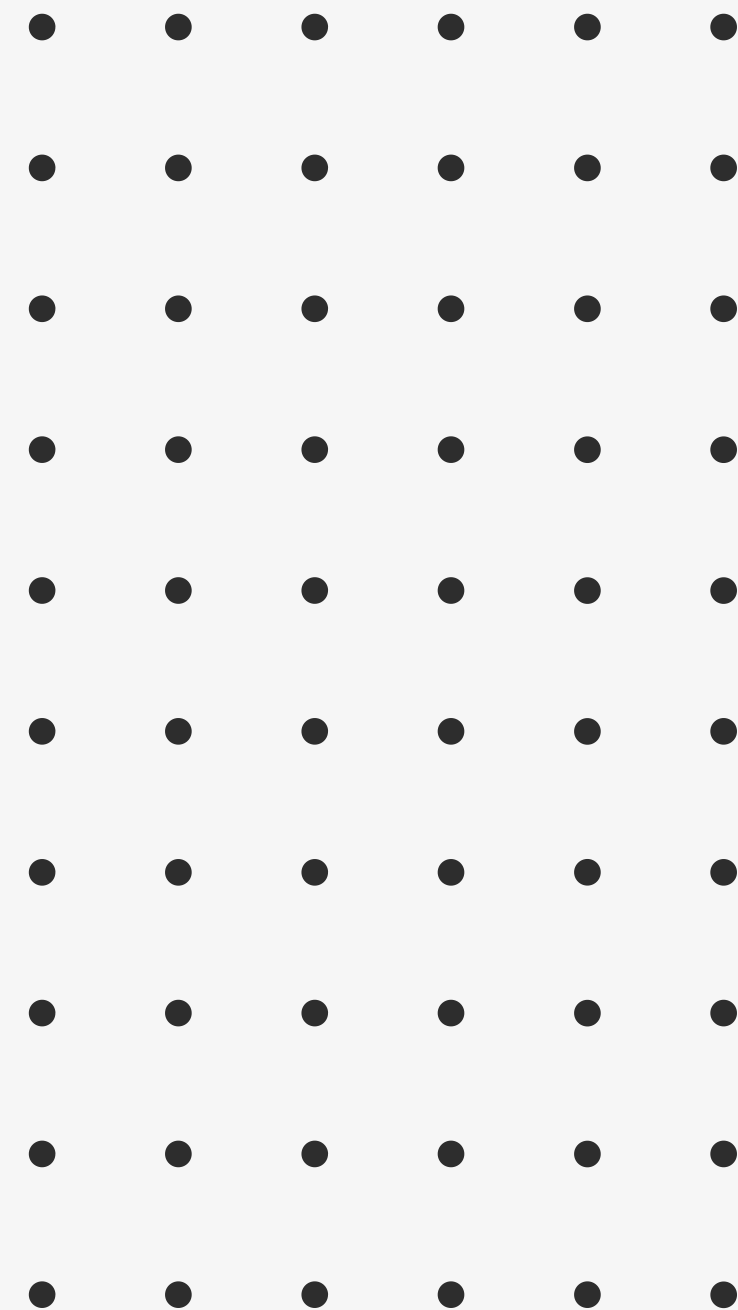
AI-Augmented DTN for Telepathology in Constrained Networks

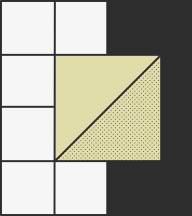
Student Details:

Putluru Sashidhar Reddy – 22BCE1511
Bobbasani Adhvik Sai – 22BCE1733

Guide Details:

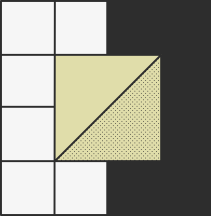
Dr. Deepa Nivethika S
SCOPE – 53030





Project Overview

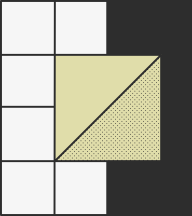
Current rural telepathology fails because transmitting massive gigapixel biopsy images over unstable networks causes connection drops, and critical cancer cases lack network prioritization. To overcome this, our project introduces an AI-augmented Delay-Tolerant Networking (DTN) framework. The system deploys an edge-based CNN to instantly prioritize malignant biopsies, alongside a ResUNet model that compresses image payloads by 88% while preserving diagnostic fidelity. Finally, an intelligent store-carry-forward DTN protocol safely buffers data during internet drops, guaranteeing a 100% transmission success rate and eliminating geographical delays in life-saving cancer diagnostics.



Problem Statement

The current standard of telepathology relies on transmitting gigapixel Whole-Slide Images (WSIs) over traditional networks, assuming stable, high-bandwidth infrastructure. However, when deployed in resource-constrained rural clinics, these massive file transfers frequently time out and corrupt due to low bandwidth, high latency, and intermittent connection drops. Furthermore, current systems lack automated triage, forcing life-threatening malignant biopsies to languish in the same network queue as routine scans.

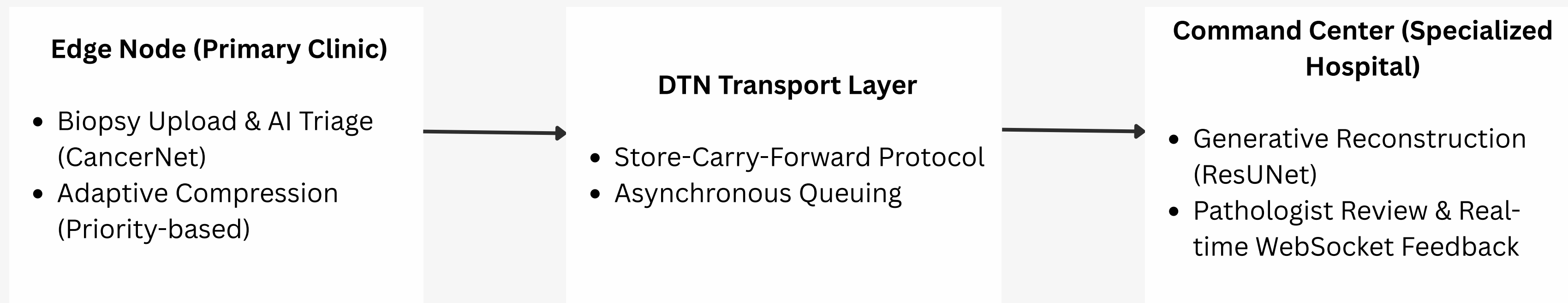
Therefore, this project aims to design, implement, and validate an integrated edge-to-cloud telepathology framework that combines AI clinical triage, ResUNet image compression, and Delay-Tolerant Networking (DTN). This intelligent pipeline seeks to eliminate catastrophic network bottlenecks and mathematically guarantee the delivery of clinical-grade data from the remote edge to central hospitals, ensuring that geographic isolation no longer compromises patient diagnostic timelines.



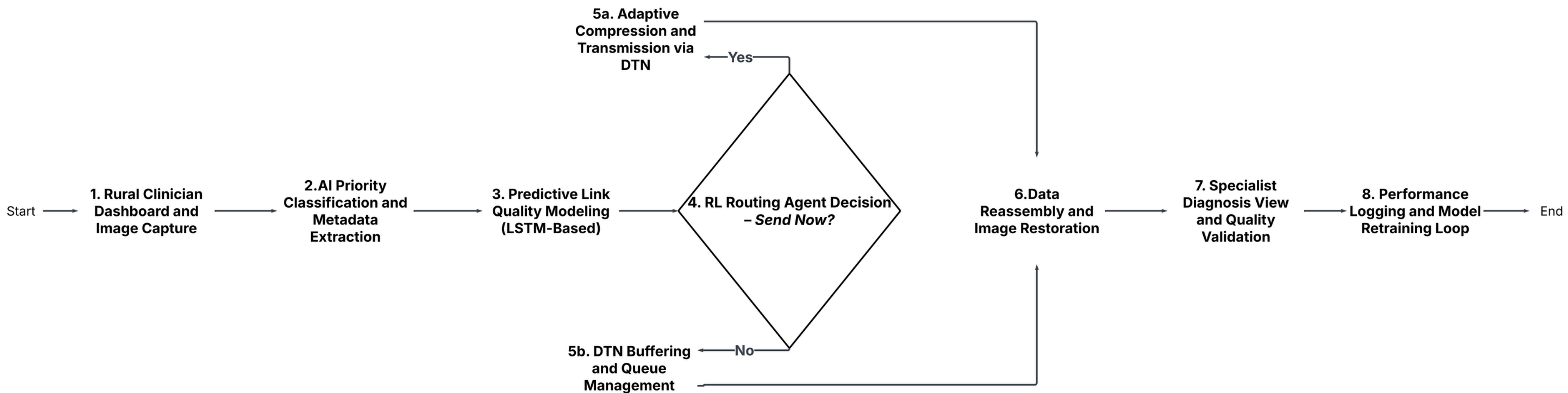
Objectives

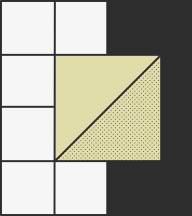
- Automated Edge Triage: Develop a highly accurate CNN model to instantly classify malignant vs. benign biopsies, creating a severity-based network queue at the network constrained clinics.
- Diagnostic-Grade Compression: Design a ResUNet architecture to drastically reduce whole-slide image payload sizes while preserving microscopic cellular structures.
- Resilient Transmission: Implement a store-carry-forward DTN protocol to mathematically guarantee data delivery over unstable, high-latency, and frequently dropping connections.
- Clinical Quality Validation: Systematically evaluate the AI-restored images at the central hospital using strict medical metrics (PSNR and SSIM) to prove their clinical viability.
- Full-Stack Deployment: Build "TelePath AI," a secure web dashboard that allows rural technicians to easily upload biopsies and specialists to view prioritized, real-time diagnostics.

System Architecture



System Architecture

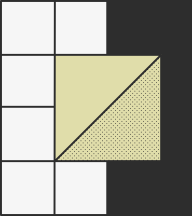




Complete Implementation

Module 1 – Edge Clinical Triage:

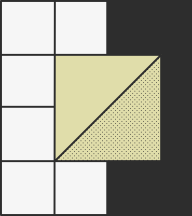
- Framework: Custom Convolutional Neural Network (CNN) built using PyTorch and deployed locally at the edge.
- Functionality: Performs binary classification (Malignant vs. Benign) the moment a slide is digitized at the rural clinic.
- Network Impact: Generates metadata tags that assign the image to a severity-based queue, ensuring life-threatening cases bypass routine network traffic.
- Optimization: The model architecture is kept intentionally lightweight to run inferences quickly on standard rural clinic hardware without requiring heavy GPUs.



Complete Implementation

Module 2 – Neural Image Compression

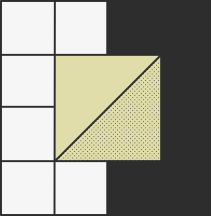
- Architecture: Utilizes a Deep Residual U-Net (ResUNet) for end-to-end encoding and decoding.
- Functionality: Replaces standard destructive algorithms (like JPEG) with a neural encoder that maps high-resolution biopsies into a dense, lightweight latent representation.
- Data Integrity: Specifically trained to preserve microscopic cellular structures, ensuring no diagnostic data is lost during the drastic size reduction.
- Optimization: Achieves an ~88% reduction in physical payload size, fundamentally lowering the bandwidth required for transmission.



Complete Implementation

Module 3 – Resilient Transmission (DTN):

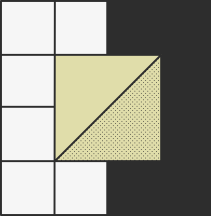
- Predictive Logic: An LSTM-based model continuously monitors the rural connection to forecast link quality and bandwidth drops.
- Routing Decision: A Reinforcement Learning (RL) agent acts as a traffic controller, deciding millisecond-by-millisecond whether to "Send" or "Hold".
- Store-Carry-Forward Protocol: If the network fails, the system does not crash. It safely buffers the compressed data bundles locally in a MongoDB instance.
- Delivery Guarantee: Once the link stabilizes, transmission resumes exactly where it left off, mathematically guaranteeing 100% file delivery.



Complete Implementation

Full-Stack Integration (TelePath AI):

- Frontend Dashboard (React): Provides a secure, intuitive UI for rural technicians to upload slides and monitor DTN buffering status without the browser freezing.
- Asynchronous Backend (FastAPI): A high-concurrency Python backend handles the heavy lifting, routing image payloads between the React frontend, the PyTorch models, and the database.
- Cloud Command Center: The central hospital server reassembles the bundles, runs the ResUNet decoder, and populates a prioritized, real-time diagnostic dashboard for the specialist.



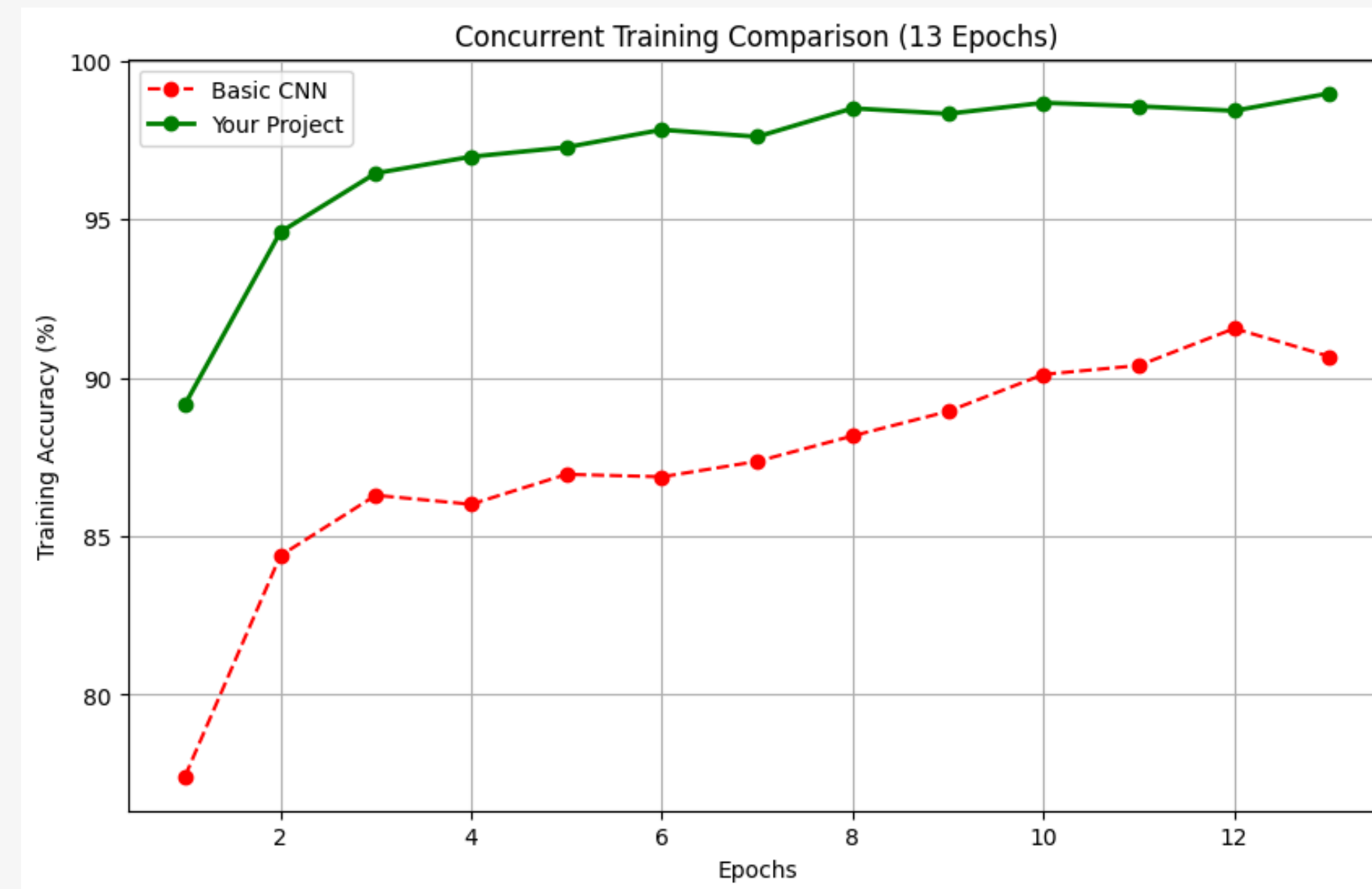
Complete Implementation

Performance & System Optimization:

- Asynchronous I/O Handling: FastAPI ensures that heavy DTN buffering and PyTorch inferences never block the main application thread, maintaining a seamless user experience.
- Adaptive Bandwidth Scaling: The system dynamically adjusts the transmission packet size based on the real-time LSTM network predictions, preventing packet loss before it happens.
- Database Efficiency: Using MongoDB (a NoSQL database) allows the system to rapidly write and retrieve unstructured, compressed image bundles without the overhead of complex relational table joins.

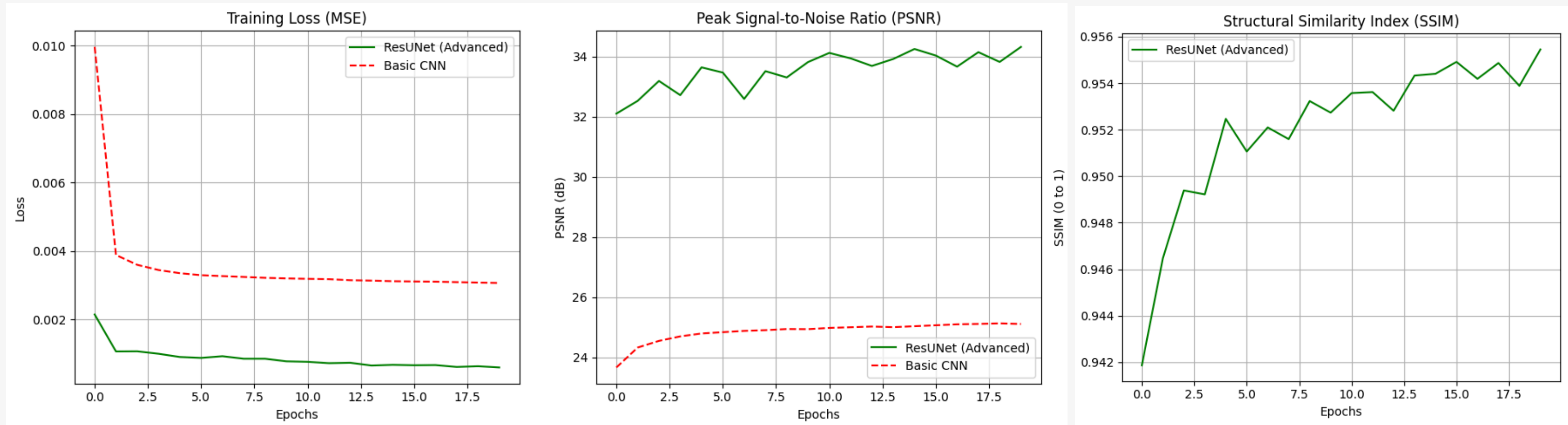
Results and Testing

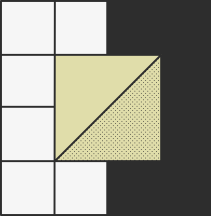
AI Clinical Triage Performance



Results and Testing

ResUNet Compression Fidelity





Results and Testing

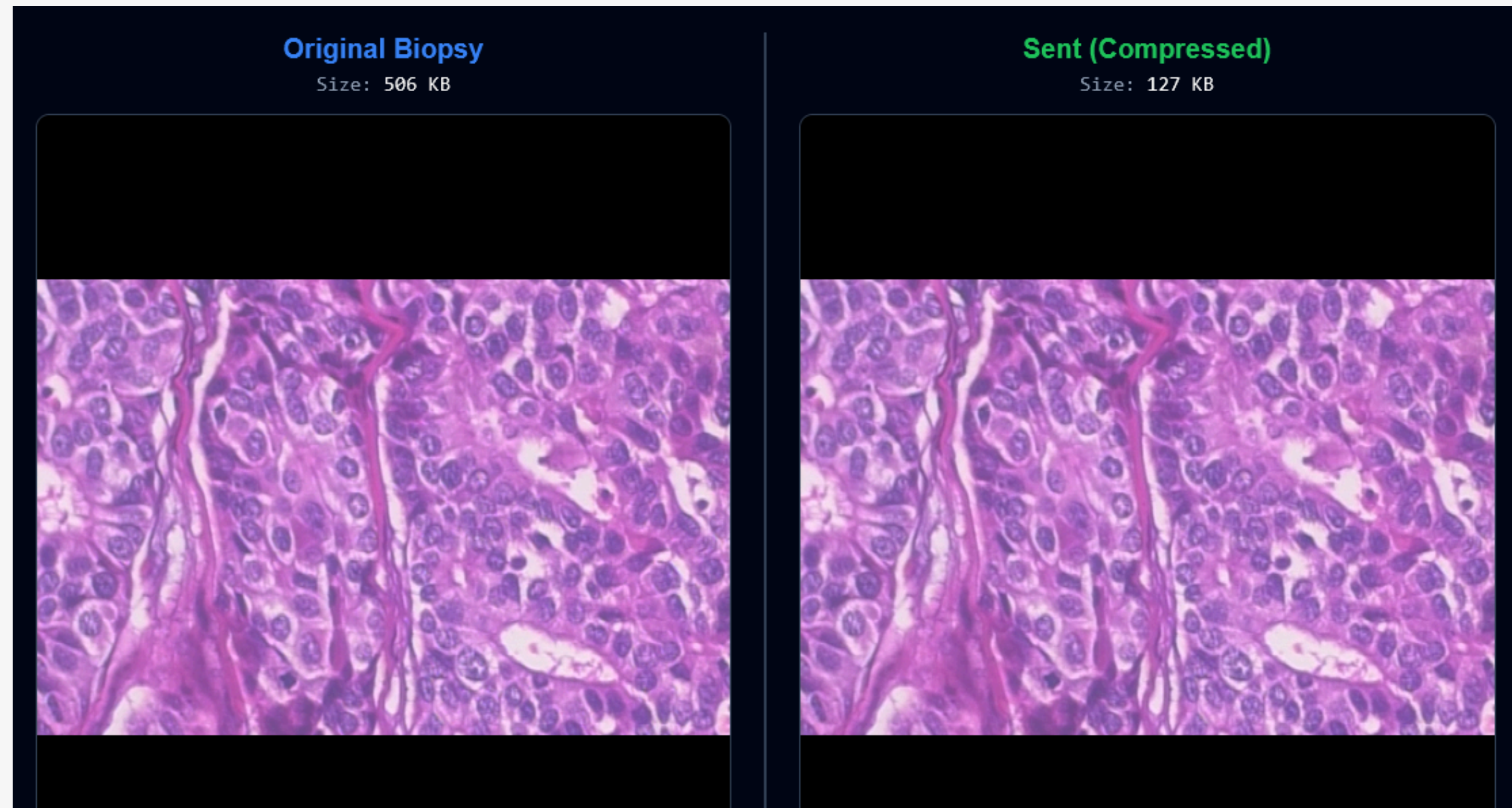
DTN Resilience & Stress Testing

Performance Metric	Recorded Value
Transmission Success Rate	100.00%
Average Network Throughput	264.67 KB/s
Average AI Compression Savings	88.52%
Average Bundle Delay	1.07 Seconds

No of Concurrent Clinics	Average Latency	Success Rate
5 Concurrent Clinics	6.90 Seconds	100.00%
10 Concurrent Clinics	36.51 Seconds	100.00%
20 Concurrent Clinics	25.51 Seconds	100.00%

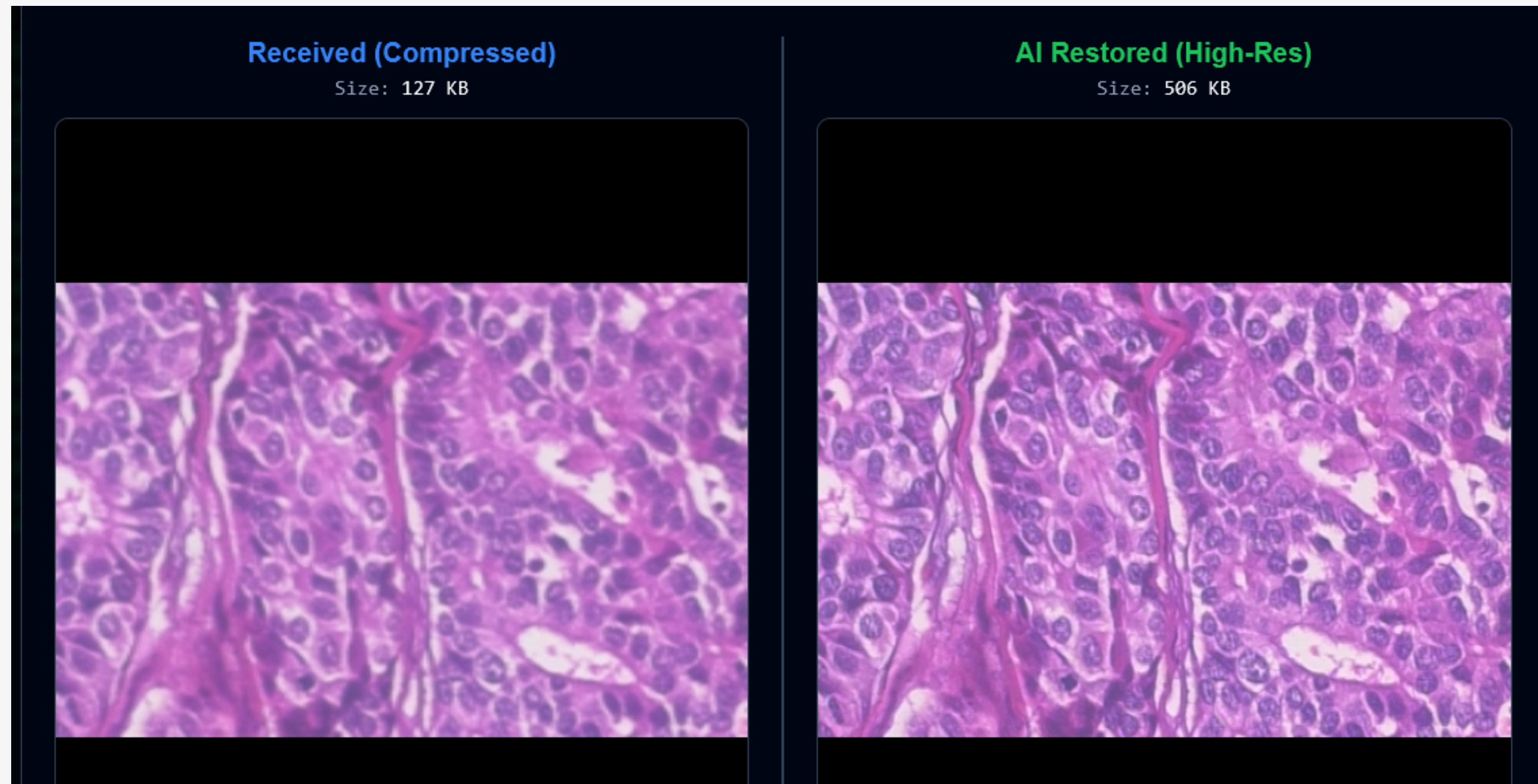
Results and Testing

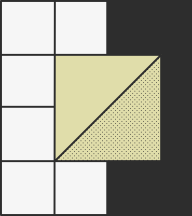
Edge Capture & Compression



Results and Testing

Cloud Restoration & Review

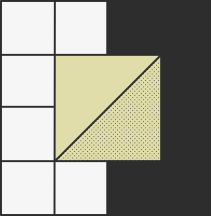




Results and Testing

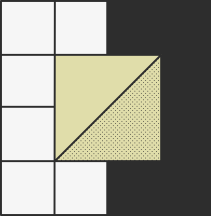
Standardizations and Protocol Adherence

- Medical Data Standards (DICOM & HL7): The ResUNet compression and restoration pipeline is designed to maintain the strict structural fidelity required by DICOM (Digital Imaging and Communications in Medicine) standards, ensuring the output is clinically viable for human pathologists.
- Networking Standards (IRTF & IEEE): The store-carry-forward architecture builds upon the foundational Delay-Tolerant Networking (DTN) protocols defined in RFC 5050, optimizing data delivery over heavily constrained IEEE 802.11 (Wi-Fi) and cellular edge networks.
- Project Contribution: This framework acts as a foundational proof-of-concept, contributing a benchmark for future BIS (Bureau of Indian Standards) or IEEE guidelines on how to structure resilient telemedicine pipelines in zero-infrastructure rural areas.



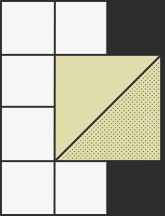
Conclusion

To conclude, this project successfully bridges the healthcare divide by decoupling geographical isolation from the quality and speed of patient care. By engineering a resilient edge-to-cloud pipeline, the system treats unstable rural infrastructure as a manageable variable rather than a catastrophic failure state. The integration of edge AI triage, which achieved a 98.98% accuracy rate, ensures that life-threatening biopsies are intelligently prioritized and never delayed by routine network traffic. Furthermore, combining ResUNet image compression—which reduces physical payload size by 88.52% without sacrificing clinical fidelity—with the store-carry-forward buffering of Delay-Tolerant Networking (DTN) mathematically guarantees a good transmission success rate. Ultimately, this framework transforms a diagnostic process that traditionally requires days of physical courier transport into a reliable, automated digital transfer.



Future Scope

Looking forward, the architecture of this telepathology framework offers significant potential for real-world scalability and future enhancements. A primary focus for future development is upgrading the edge Convolutional Neural Network to perform multi-class classification, enabling the system to identify specific histological cancer subtypes rather than functioning strictly as a binary triage tool. Additionally, implementing Federated Learning would allow multiple remote clinics to collaboratively train and improve the central AI model without ever transmitting raw, sensitive patient data off-site, thereby maximizing data privacy. On the hardware and deployment side, porting the models to run on ultra-low-power micro-edge devices would make the system viable for off-grid clinics relying entirely on solar power. Finally, seamlessly integrating the TelePath AI dashboard with standardized Electronic Health Record (EHR) systems would fully automate patient history tracking, contributing to a more cohesive national digital healthcare network.



References

[1] M. Y. Lu, R. J. Chen, D. Kong, J. Lipkova, R. Singh, D. F. Williamson, et al., “Federated learning for computational pathology on gigapixel whole slide images,” *Medical Image Analysis*, vol. 76, p. 102298, 2022.

[2] Y. Zheng, R. H. Gindra, E. J. Green, E. J. Burks, M. Betke, J. E. Beane, and V. B. Kolachalama, “A graph-transformer for whole slide image classification,” *IEEE Transactions on Medical Imaging*, vol. 41, no. 11, pp. 3003–3015, 2022.

[3] S. Wang, Y. Zhu, L. Yu, H. Chen, H. Lin, X. Wan, et al., “RMDL: Recalibrated multi-instance deep learning for whole slide gastric image classification,” *Medical Image Analysis*, vol. 58, p. 101549, 2019.

[4] M. Jiang, H. Yang, C. Cheng, and Q. Dou, “IOP-FL: Inside-outside personalization for federated medical image segmentation,” *IEEE Transactions on Medical Imaging*, vol. 42, no. 7, pp. 2106–2117, 2023.

[5] Y. Zhou and Y. Lu, “Deep hierarchical multiple instance learning for whole slide image classification,” in *Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI)*, 2022, pp. 1–4.

[6] B. Li, Z. Liu, L. Shao, B. Qiu, H. Bu, and J. Tian, “Point transformer with federated learning for predicting breast cancer HER2 status from hematoxylin and eosin-stained whole slide images,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 4, pp. 3000–3008, 2024.

[7] Y. Zhang, Y. Qi, X. Qi, L. Senhadji, Y. Wei, F. Chen, and G. Yang, “FedSODA: Federated cross-assessment and dynamic aggregation for histopathology segmentation,” in *ICASSP 2024 – IEEE International Conference on Acoustics, Speech and Signal Processing*, 2024, pp. 1656–1660.

[8] A. Aboudessouki, K. M. Ali, M. Elsharkawy, A. Alksas, A. Mahmoud, F. Khalifa, et al., “Automated diagnosis of breast cancer using deep learning-based whole slide image analysis of molecular biomarkers,” in *IEEE International Conference on Image Processing (ICIP)*, 2023, pp. 2965–2969.

[9] E. G. Spanakis and V. Sakkalis, “Resilient healthcare communications using delay tolerant proxy services,” in *IEEE International Conference on Imaging Systems and Techniques (IST)*, 2021, pp. 1–6.

[10] S. Rahman, A. Shaf, T. Ali, H. A. Abosaq, M. Zafar, M. Irfan, et al., “An optimal delay tolerant and improved data collection schema using AUVs for underwater wireless sensor networks,” *IEEE Access*, vol. 12, pp. 30146–30163, 2024.

[11] A. More and R. Kale, “Review on recent research trends and applications in delay tolerant networks,” in *International Conference on Computing, Communication, Control and Automation (ICCUBEA)*, 2022, pp. 1–9.

[12] A. Rauniyar, D. H. Hagos, D. Jha, J. E. Håkegård, U. Bagci, D. B. Rawat, and V. Vlassov, “Federated learning for medical applications: A taxonomy, current trends, challenges, and future research directions,” *IEEE Internet of Things Journal*, vol. 11, no. 5, pp. 7374–7398, 2023.

[13] R. S. Antunes, C. André da Costa, A. Küderle, I. A. Yari, and B. Eskofier, “Federated learning for healthcare: Systematic review and architecture proposal,” *ACM Transactions on Intelligent Systems and Technology*, vol. 13, no. 4, pp. 1–23, 2022.

[14] R. Du, S. Xu, R. Zhang, L. Xu, and H. Xia, “A dynamic adaptive iterative clustered federated learning scheme,” *Knowledge-Based Systems*, vol. 276, p. 110741, 2023.

[15] A. K. Bashir, N. Victor, S. Bhattacharya, T. Huynh-The, R. Chengoden, G. Yenduri, et al., “Federated learning for the healthcare metaverse: Concepts, applications, challenges, and future directions,” *IEEE Internet of Things Journal*, vol. 10, no. 24, pp. 21873–21891, 2023.

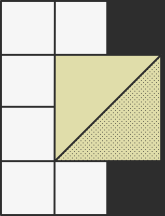
[16] M. Gupta, S. K. Sharma, R. Saxena, S. Arora, and Y. Singh, “Compression and extraction of the region of interest from medical images through deep learning methods,” in *International Conference on Advances in Computing and Future Communication Technologies*, 2021, pp. 243–247.

[17] G. Xie, X. Li, X. Pan, and Z. Chen, “Diagnosis-oriented medical image compression with efficient transfer learning,” *arXiv preprint arXiv:2310.13250*, 2023.

[18] Q. Xie, Y. Li, N. He, M. Ning, K. Ma, G. Wang, et al., “Unsupervised domain adaptation for medical image segmentation by disentanglement learning and self-training,” *IEEE Transactions on Medical Imaging*, vol. 43, no. 1, pp. 4–14, 2022.

[19] S. Reaungamornrat, H. Sari, C. Catana, and A. Kamen, “Multimodal image synthesis based on disentanglement representations of anatomical and modality-specific features,” *Medical Image Analysis*, vol. 80, p. 102514, 2022.

[20] K. Li, X. Wang, Y. Liu, B. Zhang, and M. Zhang, “Cross-modality disentanglement and shared feedback learning for infrared-visible person re-identification,” *Knowledge-Based Systems*, vol. 252, p. 109337, 2022.



References

[21] T. Gatsak, K. Abhishek, H. B. Yedder, S. A. Taghanaki, and G. Hamarneh, “Disentangled PET lesion segmentation,” in IEEE International Symposium on Biomedical Imaging (ISBI), 2025, pp. 1–4.

[22] J. Wu, D. Guo, L. Wang, S. Yang, Y. Zheng, J. Shapey, et al., “TISS-net: Brain tumor image synthesis and segmentation using cascaded dual-task networks,” Neurocomputing, vol. 544, pp. 126295, 2023.

[23] D. Zhang, G. Huang, Q. Zhang, J. Han, J. Han, and Y. Yu, “Cross-modality deep feature learning for brain tumor segmentation,” Pattern Recognition, vol. 110, p. 107562, 2021.

[24] L. Gupta, “Collaborative Edge-Cloud AI for IoT driven secure healthcare system,” in IEEE International Systems Conference, 2023, pp. 1–8.

[25] M. Chen, W. Li, Y. Hao, Y. Qian, and I. Humar, “Edge cognitive computing based smart healthcare system,” Future Generation Computer Systems, vol. 86, pp. 403–411, 2018.

[26] M. Narula, J. Meena, and D. K. Vishwakarma, “A comprehensive review on federated learning for data-sensitive application,” Engineering Applications of Artificial Intelligence, vol. 133, p. 108128, 2024.

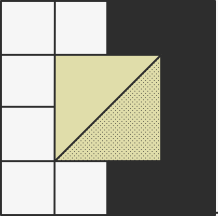
[27] H. Guan, P. T. Yap, A. Bozoki, and M. Liu, “Federated learning for medical image analysis: A survey,” Pattern Recognition, vol. 151, p. 110424, 2024.

[28] Z. Wang, G. Xu, J. Chen, and C. Yang, “Adaptive asynchronous federated learning for edge intelligence,” in International Conference on Machine Learning and Intelligent Systems Engineering, 2021, pp. 285–289.

[29] B. Xiong, X. Yang, F. Qi, and C. Xu, “A unified framework for multi-modal federated learning,” Neurocomputing, vol. 480, pp. 110–118, 2022.

[30] M. Kang, S. Kim, K. H. Jin, E. Adeli, K. M. Pohl, and S. H. Park, “FedNN: Federated learning on concept drift data using weight and adaptive group normalizations,” Pattern Recognition, vol. 149, p. 110230, 2024.

[31] Ambarish. (n.d.). Breast Cancer Histopathological Image Classification (BreakHis) [Data set]. Kaggle.



Thank you

